the little

PACKET

who wanted to see the world

a bedtime story for geeks of all ages

once upon a time...

someone typed **cats** into a search box and pressed enter.

and a little packet was born. it had a header, a payload, and a time-to-live of **64** hops.

it wanted to see the world.

first stop: DNS

"excuse me," said the packet. "where does search.example live?"

the wise old resolver checked her notes. "ah. **203.0.113.7**. off you go, dear."

routers. so many routers.

each one squinted at the address, pointed vaguely east, and said *"that way."*

TTL 63 ... 62 ... 61 ...

60 ...

(the packet tried not to think about what happens at zero.)

under the sea

the packet rode a cable along the bottom of the ocean, as fast as light in glass.

a shark swam over for a nibble. (sharks really do this.)

knock knock

at last, the server! the packet knocked politely.

packet: **SYN?**
server: **SYN-ACK!**
packet: **ACK.**

and just like that, they were friends.

made it!

the little packet handed over its one word: **cats**.

then it checked its stamp. **TTL 49.** zero never even came close.

it had seen the world. (well, the internet.)

the end?

the server thought for three whole milliseconds, then sent back 4,096 packets of cats. (servers are like that.)

one of them took the long way home and ran out of hops.

don't worry. it'll get sent again. *it always does.*

tiny glossary
DNS — the internet's phone book
router — a very confident signpost
TTL — hops left before giving up
ACK — "got it!"

cut along the dashes